

# Project

**Topic:** Data Cleaning, Analysis, and Visualization

Project by Maria Besedina

## The project consisted of the following stages:

### Data Cleaning

- Processed and prepared CRM data using Python
- Removed duplicates, missing values, and incorrect entries

### Exploratory Data Analysis (EDA)

- Analyzed data structure and feature distributions
- Identified patterns and anomalies in lead behavior, manager performance, and their impact on deal success

### Data Visualization

- Created an interactive dashboard in Microsoft Power BI
- Visualized key metrics and business insights

# Stage 1: Data Cleaning

**Tools:** Python (*pandas*, *numpy*)

## **Project Description:**

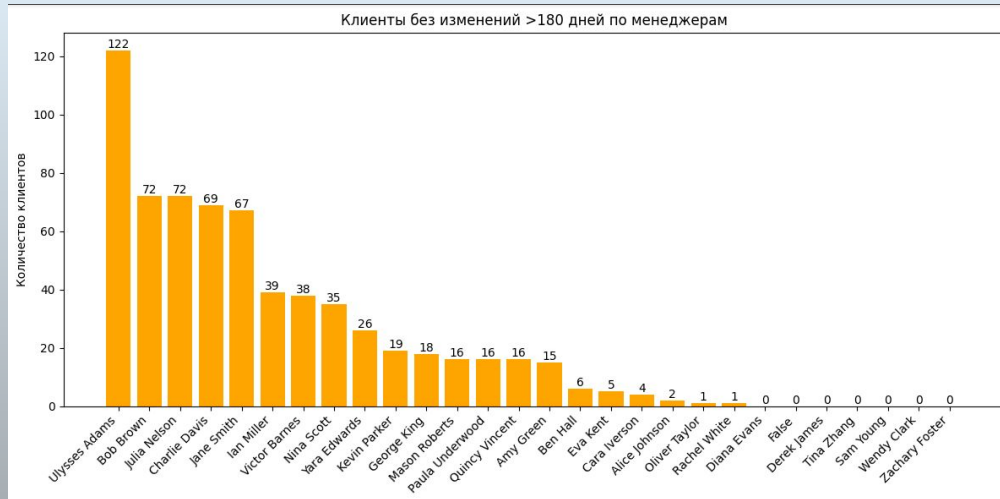
- The raw CRM data contained numerous unnecessary characters, missing values, and inconsistent formatting.
- Text cleaning was performed, including the removal of extra symbols and spaces, as well as the standardization of naming conventions and data formats.
- Due to limited data availability, row deletion was minimized whenever possible, with preference given to data modification and correction instead.
- Data records from multiple tables were matched and consolidated to improve data integrity and consistency.

**Result:** Prepared a clean and structured dataset suitable for further analysis and visualization.

# Key Insights from the Analysis

## Identified Patterns:

- Analysis of client distribution by manager revealed that some clients, despite being registered long ago, demonstrated no recent activity
- The **modification\_time** field indicated that some records had not been updated for over 180 days
- These clients showed no engagement and did not progress through the sales funnel.



## Hypothesis / Recommendation:

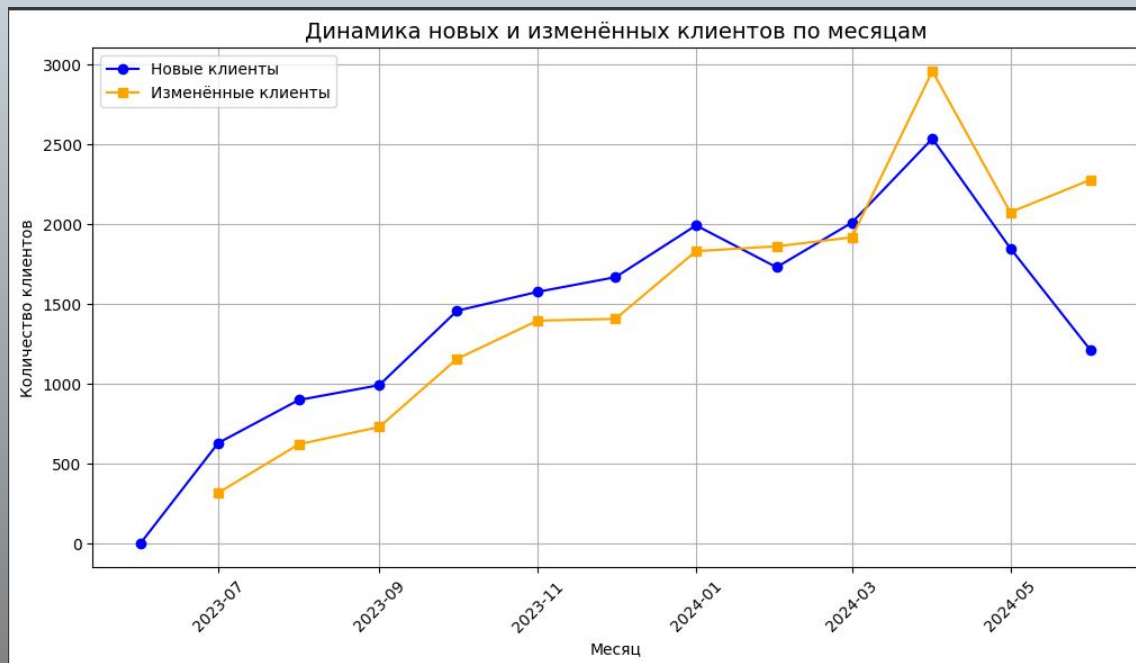
- **Recommendation: Initiate re-engagement campaigns to remind these clients about available courses and offerings**
- If re-engagement is unsuccessful, clients should be archived to maintain data quality and ensure accurate reporting.

# Analysis of Seasonal Activity

## Analysis Overview:

The analysis involved visualizing the monthly distribution of leads and calls

Both metrics were visualized on a single chart, revealing a positive correlation between lead volume and call activity.



## Key Observations and Patterns:

**The analysis revealed a positive relationship between call volume and lead growth.**

**Clear seasonality in activity was observed.**

- A peak in activity was observed in December, driven by clients aiming to complete transactions before year-end.
- A comparable peak in activity was observed in May, prior to the summer holiday period.
- After holiday periods, engagement in training typically rebounds, likely driven by renewed motivation following a break.

## Key Conclusion:

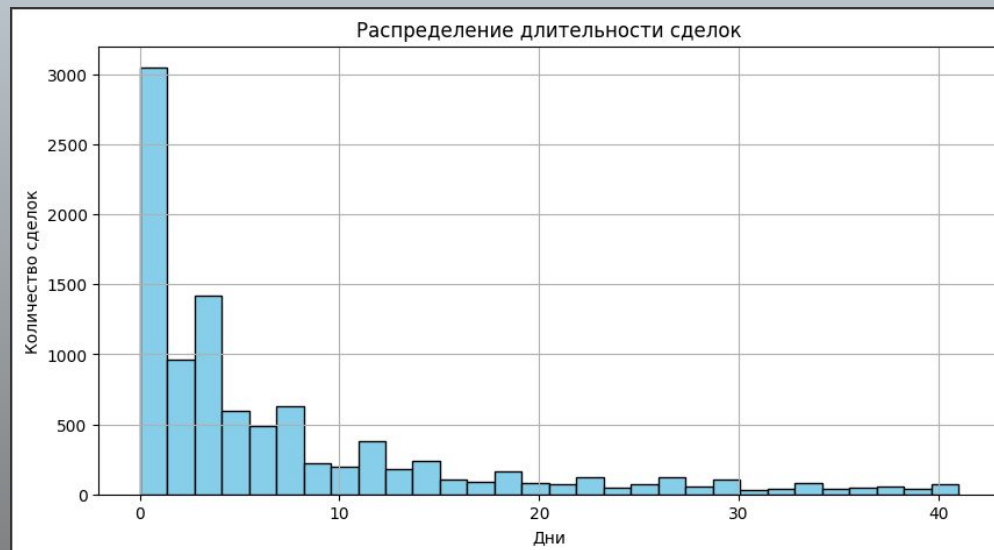
Client activity shows clear seasonality, which should be considered when planning marketing campaigns and allocating sales team workload.



# Deal Duration Distribution

## Key Findings:

- The average deal duration ranges from 3–4 days, with a median of approximately 1 day.
- The majority of deals are closed within 1–2 days, with durations exceeding one week being rare.
- Prior to data cleaning, anomalies were detected, including invalid negative and extreme values ranging from -358 to 334 days.
- After filtering, the data range became realistic, and the distribution appeared right-skewed with a long tail.





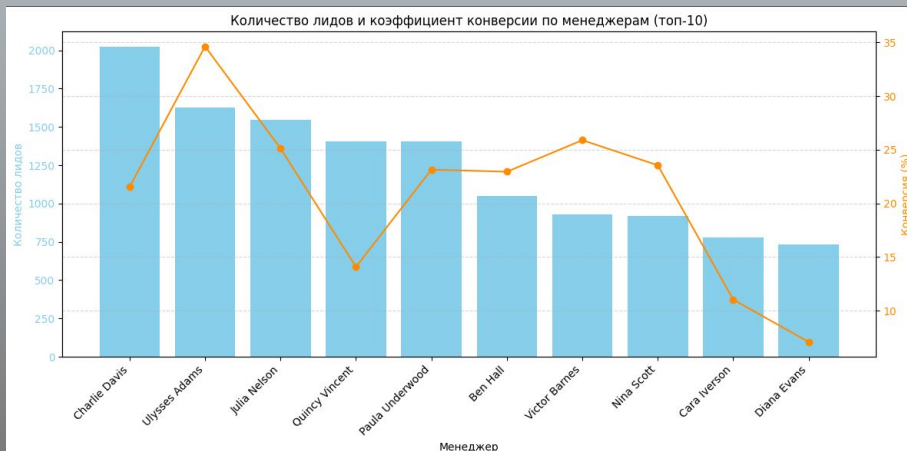
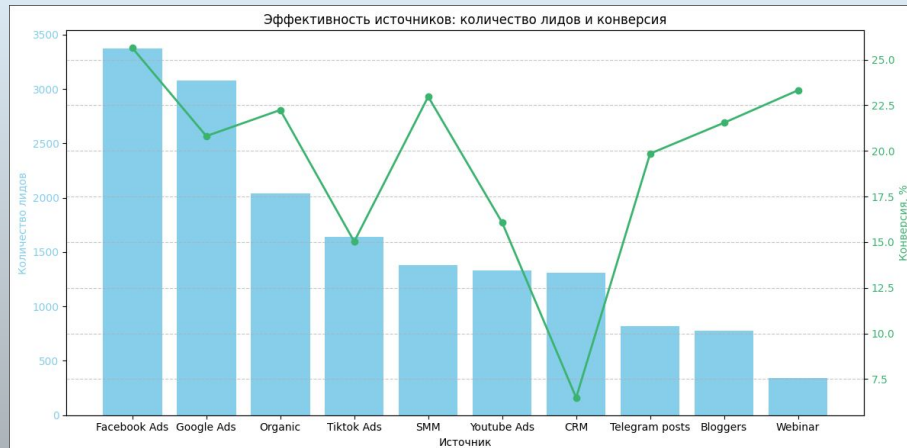
## Conclusion: Lead and Conversion Analysis

Analysis across managers, campaigns, and sources revealed that a higher lead volume does not guarantee higher conversion rates.

**This discrepancy can be attributed to several factors:**

- such as audience characteristics and lead quality;
- **as well as human factors, such as differences in managers' experience and approaches;**
- and the characteristics of advertising channels that may attract lower-intent clients.

Overall, the results indicate that performance evaluation should account not only for lead volume but also for lead processing efficiency.

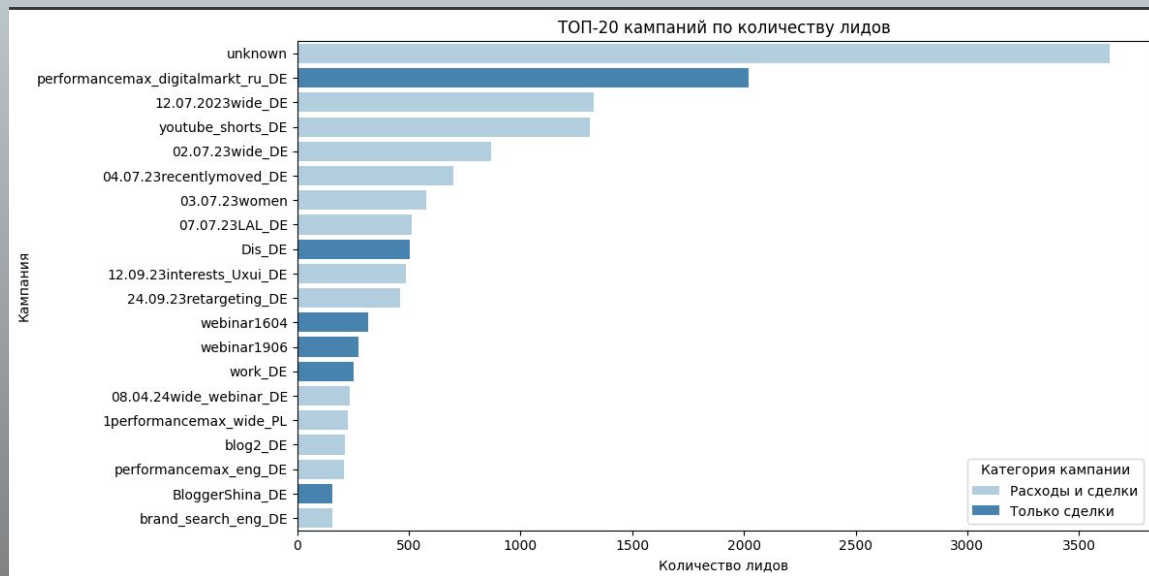


# Analysis of Advertising Campaign Performance

## Description:

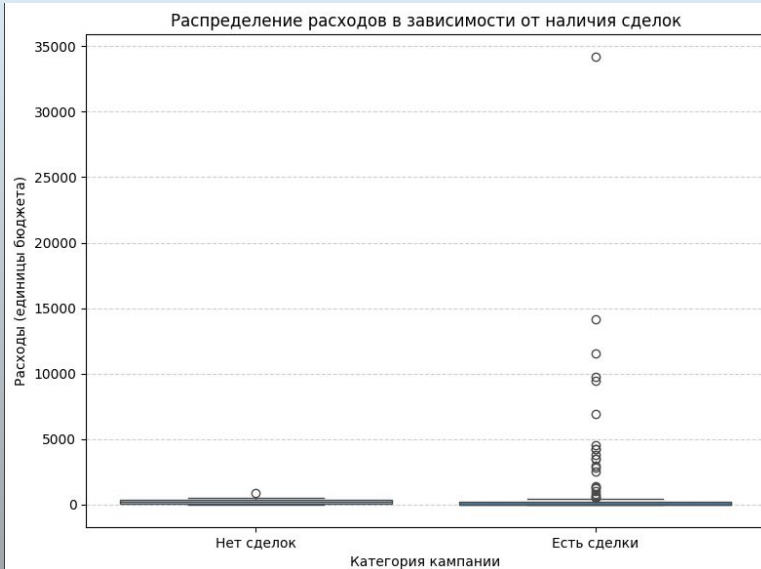
All campaigns were segmented into distinct groups during the analysis:

1. Spending and Deals — campaigns with both costs and generated leads.
2. Deals Only — campaigns that generate deals with no associated spend.



## Results:

- Most campaigns are classified as “Spend and Deals,” indicating that while advertising generates results, it requires budget investment
- The second-largest group is “Deals Only,” consisting of zero-spend campaigns that still generate a meaningful number of successful deals.



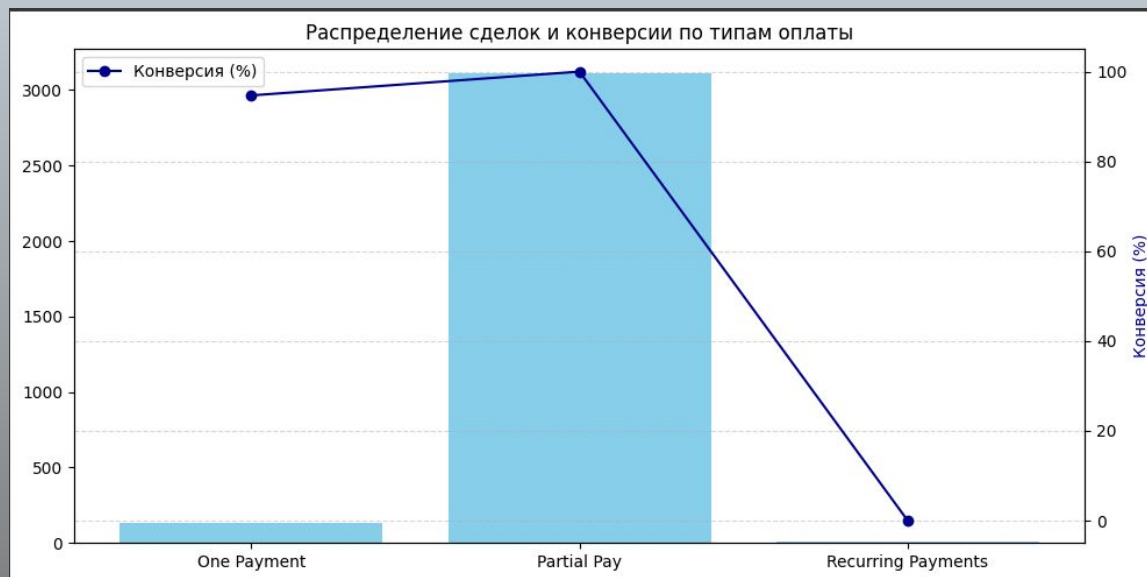
## Conclusion:

While advertising investment remains important, the analysis indicates that some deals can be generated without direct spend. This suggests an opportunity to optimize marketing costs by focusing on high-performing, cost-efficient campaigns and sources.

# Deal Distribution by Payment Type

## Analysis Overview:

The analysis examined deal distribution and conversion rates across payment types, identifying the most commonly used payment methods.



## Conclusions:

- The majority of clients prefer installment payments (Partial Pay), making it the most frequently used option.
- One Pay (single payment) is also widely used, especially for smaller transaction amounts.
- Recurring payments (Recurring Pay) are rare and are not linked to successful deals.

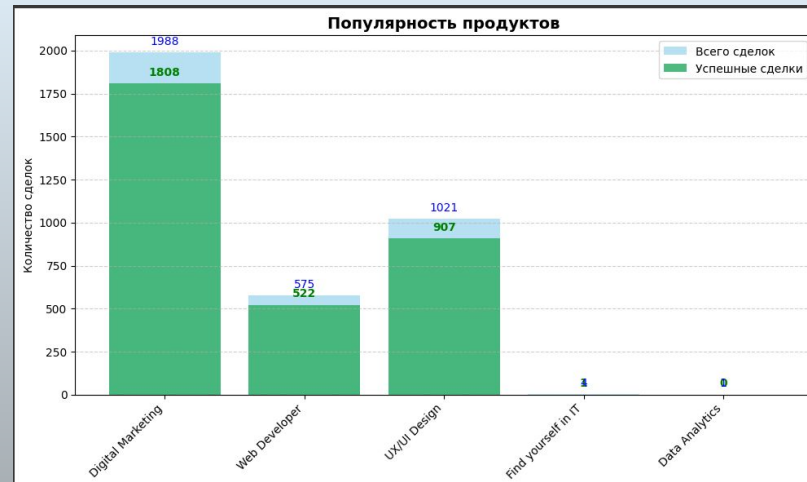
### Conclusion:

The majority of sales are processed via installment and single payments.

## Analysis of Product Demand

### Results:

- **Digital Marketing** is the most in-demand product.
- Data Analytics and “Find Yourself in IT” generate fewer deals compared to other courses. .
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|   | Product             | first_deal          | last_deal  | days_active |
|---|---------------------|---------------------|------------|-------------|
| 3 | No Cours            | 2023-07-03 17:03:00 | 2024-12-11 | 526         |
| 1 | Digital Marketing   | 2023-07-04 10:11:00 | 2024-09-25 | 448         |
| 5 | Web Developer       | 2023-07-18 16:51:00 | 2024-06-21 | 338         |
| 4 | UX/UI Design        | 2023-07-19 09:01:00 | 2024-06-20 | 336         |
| 2 | Find yourself in IT | 2023-11-17 17:20:00 | 2024-05-11 | 175         |
| 0 | Data Analytics      | 2023-08-01 18:06:00 | 2023-08-01 | -1          |

## Product Demand Analysis

### Recommendation:

- Enhance marketing efforts and integrate these products into key promotional campaigns
- This is expected to improve brand awareness and accelerate user acquisition for new courses

## Study Format Analysis (Daytime vs Evening)

### Results:

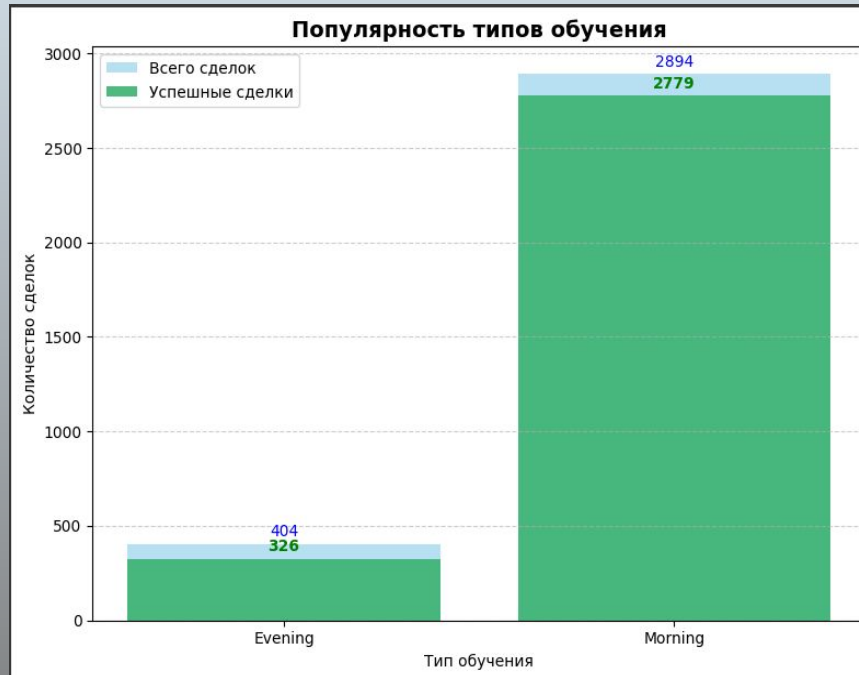
- The analysis revealed significantly higher demand for daytime programs compared to evening ones, although evening courses also remain in demand.
- A Temporal analysis confirmed that both formats were launched at the same time, ruling out novelty as a factor in demand differences .
- The most likely driver is audience preference, with daytime study being more convenient for students and non-full-time workers .

|   | Education Type | first_deal          | last_deal  | days_active |
|---|----------------|---------------------|------------|-------------|
| 2 | No Cours       | 2023-07-03 20:17:00 | 2024-12-11 | 526         |
| 1 | Morning        | 2023-07-05 18:06:00 | 2024-09-25 | 447         |
| 0 | Evening        | 2023-07-19 11:55:00 | 2024-06-20 | 336         |



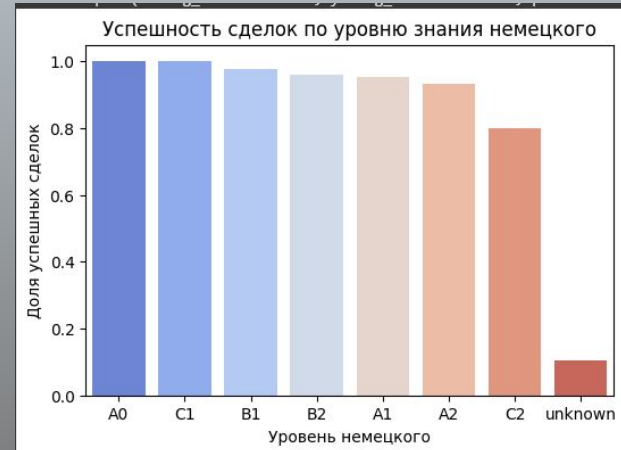
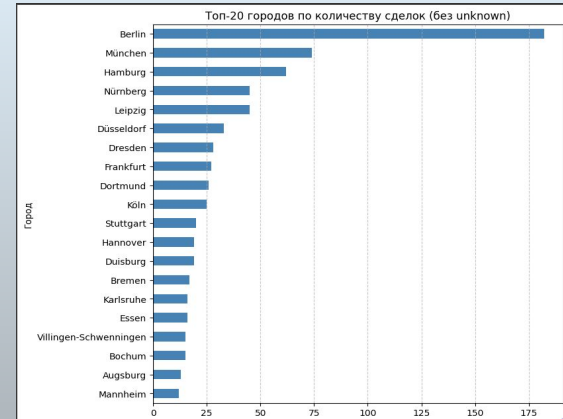
## Conclusions:

- Evening courses demonstrate stable demand
- It is recommended to maintain both formats, with a stronger marketing focus on daytime programs and gradual expansion of evening course promotion as a flexible option for working clients.



## Customer Geography:

- The top 20 cities by deal volume are predominantly German, with a notable share of international locations .
- This suggests strong potential for international expansion and justifies targeting foreign audiences .
- The analysis revealed no impact of German language proficiency on deal success, indicating relevance for both German-speaking and non-German-speaking clients and supporting international positioning .



## Projekt Summary

- CRM data cleaning and analysis successfully completed.
- Key patterns were identified:
  - Fast deal closures and seasonal activity spikes
  - Leading products and traffic sources
  - Opportunities to optimize advertising spend
- Recommendations for course promotion and development were formulated.

**All key results were visualized in an interactive dashboard built in Power BI .**

€ 133,005

Total Spend

21538

Total Deals

Data Analytics

Top Product

25%

Call2Deal Conv

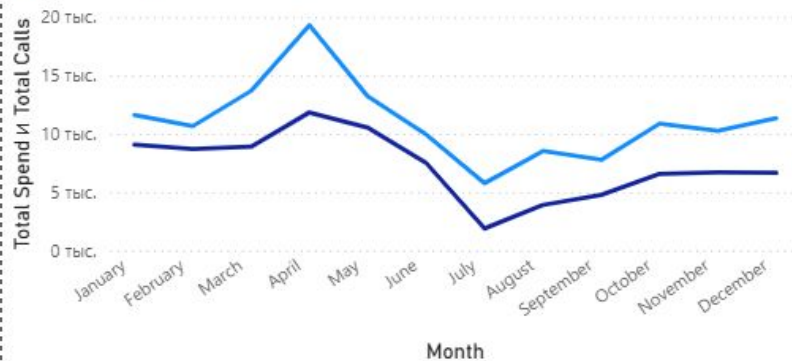


€ 3,955,112

Total Revenue

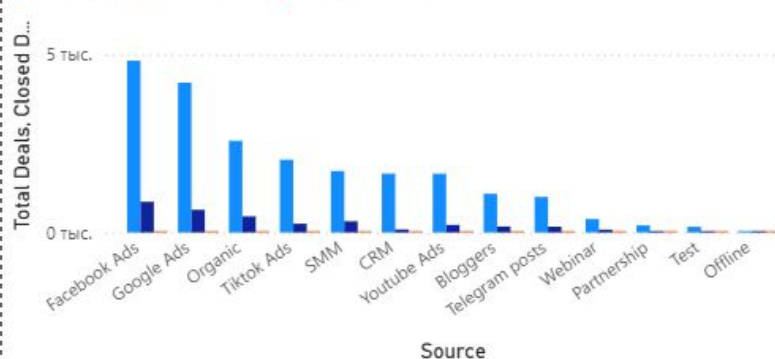
Total Spend и Total Calls по Month

● Total Spend ● Total Calls

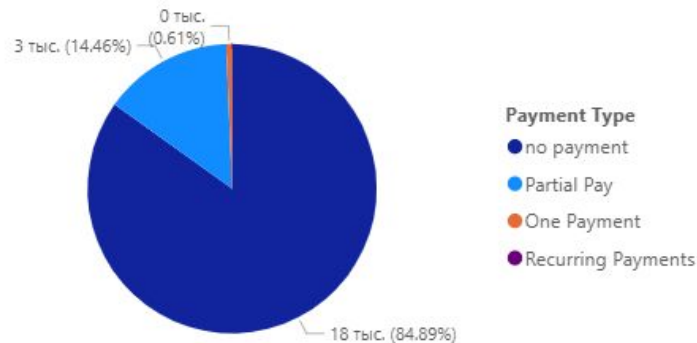


Total Deals, Closed Deals и Call2Deal Conversion по Source

● Total Deals ● Closed Deals ● Call2Deal Conversion



Total Deals по Payment Type



Deal Owner Name Total Deals Closed Deals Total Revenue Call2Deal Conversion

| Deal Owner Name | Total Deals  | Closed Deals | Total Revenue  | Call2Deal Conversion |
|-----------------|--------------|--------------|----------------|----------------------|
| Alice Johnson   | 36           | 1            | 0              | 0.00                 |
| Amy Green       | 66           | 0            | 0              | 0.00                 |
| Ben Hall        | 1345         | 241          | 241700         | 0.02                 |
| Bob Brown       | 331          | 2            | 950            | 0.00                 |
| Cara Iverson    | 1056         | 86           | 688400         | 0.01                 |
| Charlie Davis   | 2956         | 435          | 445600         | 0.03                 |
| Diana Evans     | 1011         | 52           | 50450          | 0.01                 |
| Eva Kent        | 459          | 62           | 65200          | 0.01                 |
| George King     | 107          | 4            | 2900           | 0.00                 |
| Ian Miller      | 489          | 33           | 33050          | 0.01                 |
| <b>Bcero</b>    | <b>21538</b> | <b>3262</b>  | <b>3955112</b> | <b>0.25</b>          |

City

Bce

Education Type

☐ Evening☐ Morning

Payment Type

Bce

Date

3/17/2023

12/16/2025

Campaign

☐ 01.04.23women\_PL☐ 02.05.24test\_DE☐ 02.07.23wide\_DE☐ 02.08.23interests\_DE☐ 03.07.23women☐ 04.07.23recentlymov...☐ 05.07.23interests\_DE☐ 05.09.2023wide\_DE

